

**A REPORT
ON
Intensive AI GPU Summer Internship
Program (Summer 2026)**

Submitted by

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Under the guidance of,
**Ms. Shet Reshma Prakash
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in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE & TECHNOLOGY
(Artificial Intelligence and Machine Learning)**

AT



PRESIDENCY UNIVERSITY

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PRESIDENCY UNIVERSITY

**PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND
ENGINEERING**



This is to certify that the report of **CSE7000 – Internship** on
Intensive AI GPU Summer Internship Program (Summer 2026)
being submitted by

M SHYAM bearing roll number **20241CSG0089** in partial fulfillment
of the requirement for the award of the degree of Bachelor of
Technology in **Computer Science and Technology (Artificial
Intelligence and Machine Learning)** is a bonafide work carried out
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DECLARATION

I hereby declare that the work, which is being presented in the report entitled **Intensive AI GPU Summer Internship Program (Summer 2026)** in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Technology (Artificial Intelligence and Machine Learning)** , is a record of my own investigations carried under the guidance of, **Ms. Shet Reshma Prakash** and **Mr. Sachin Kumar** , Designation, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru.

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

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ABSTRACT

Software development teams rely on code reviews to improve code quality, maintain coding standards, and identify defects before deployment. However, conventional code review tools and AI-based assistants generally operate in a stateless manner, providing generic feedback without considering a developer's previous coding patterns or recurring mistakes. This often results in repetitive suggestions, inconsistent reviews, and limited personalization.

This internship project presents the design and implementation of an AI-Powered Code Review Agent with Agentic Memory, an intelligent code review system capable of learning from historical reviews. The system integrates Large Language Models with a persistent memory layer to provide personalized and context-aware code review recommendations. Unlike traditional review assistants, the proposed system stores previous review interactions and utilizes them to adapt future responses according to the developer's coding history.

The application is developed using FastAPI for the backend and React for the frontend. The review engine is powered by the Groq API using the Qwen3-32B large language model, while persistent memory is implemented using Hindsight Memory. The application supports multiple programming languages including Python, Java, JavaScript, C, C++, Go, Rust, and TypeScript. A modern web interface with the Monaco Code Editor enables users to submit code, receive AI-generated reviews, and visualize historical improvements.

The developed system demonstrates how persistent memory can transform AI-assisted code review from a generic recommendation system into an adaptive intelligent assistant capable of recognizing developer-specific coding patterns, improving review quality, and reducing repetitive feedback. The project highlights the practical application of Artificial Intelligence, Large Language Models, cloud-based memory systems, and modern web technologies in software engineering.

INTRODUCTION

Software development involves continuous code writing, testing, debugging, and review. Code review is an important part of this process because it helps identify errors, improve code quality, maintain coding standards, and make software easier to maintain. However, traditional code review depends heavily on manual effort and can become time-consuming when reviewing large amounts of source code.

Recent developments in Artificial Intelligence (AI) and Large Language Models (LLMs) have made it possible to automate several software engineering activities, including code analysis and review. AI-based code review systems can analyze source code and provide suggestions related to errors, code quality, performance, security, and best practices. However, a major limitation of many such systems is that they treat each review as an independent interaction and do not retain useful knowledge from previous reviews.

The AI Code Review Agent with Hindsight Memory addresses this limitation by combining an AI-based code review system with persistent memory. The system can retain information from previous code reviews and use relevant historical information when analyzing new code. This allows the system to identify recurring coding issues and provide more personalized feedback instead of repeatedly generating completely generic suggestions.

The project uses a FastAPI-based backend for handling application logic and API requests, while a React-based frontend provides the user interface for submitting and reviewing code. The AI review functionality is powered by a Large Language Model through the Groq API, and Hindsight Memory is used as the persistent memory layer. The application provides features such as code review, review history, memory-based analysis, and identification of recurring coding issues.

The project demonstrates the practical application of Artificial Intelligence, Large Language Models, persistent memory, and modern web technologies in software engineering. By combining code analysis with historical context, the system aims to provide a more adaptive and useful code review experience for developers.

2. OBJECTIVES

The main objective of the project is to develop an AI-powered code review agent with persistent memory that can analyze source code and provide intelligent, personalized, and context-aware feedback.

The specific objectives are:

To develop an AI-powered automated code review system that analyzes source code and generates meaningful review suggestions.

To integrate a Large Language Model for identifying coding issues and providing recommendations for improving code quality.

To implement persistent memory using Hindsight Memory so that information from previous code reviews can be retained and retrieved.

To identify recurring coding issues by using information from previous review interactions.

To provide personalized code review feedback by incorporating relevant historical review context into new reviews.

To maintain a history of code reviews so that developers can access and analyze their previous review results.

To provide a user-friendly web interface for submitting source code and viewing AI-generated review results.

To develop a modular and scalable system architecture using a FastAPI backend and React frontend.

To demonstrate the application of AI and persistent memory in software engineering, particularly in automated code review.

To support continuous improvement in coding practices by helping developers recognize repeated mistakes and understand areas requiring improvement.

CONCLUSION

The AI Code Review Agent with Hindsight Memory successfully demonstrates the integration of Artificial Intelligence, Large Language Models, and persistent memory to create an intelligent code review system. The project addresses a major limitation of conventional AI-based code review tools, which generally provide independent and generic feedback without retaining knowledge from previous interactions.

The developed system combines a React frontend, FastAPI backend, Groq API, and Hindsight Memory to provide an interactive and memory-enabled code

review experience. The system can analyze submitted source code, generate AI-based feedback, store relevant review information, and retrieve previous knowledge to provide more context-aware recommendations.